US 20250287678A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0287678 A1
(43) Pub. Date: Sep. 11, 2025
Hsu et al.

(54) **MANUFACTURING METHOD OF SEMICONDUCTOR INTEGRATED STRUCTURE HAVING HIGH/LOW VOLTAGE DEVICES AND CAPACITOR**

(71) Applicant: **Richtek Technology Corporation**, Zhubei City (TW)

(72) Inventors: **Yu-Wei Hsu**, Tainan (TW); **Han-Chung Tai**, Hsinchu (TW); **Chun-Lung Chang**, Hsinchu (TW)

(21) Appl. No.: **18/666,904**

(22) Filed: **May 17, 2024**

(30) **Foreign Application Priority Data**

Mar. 8, 2024 (TW) 113108618

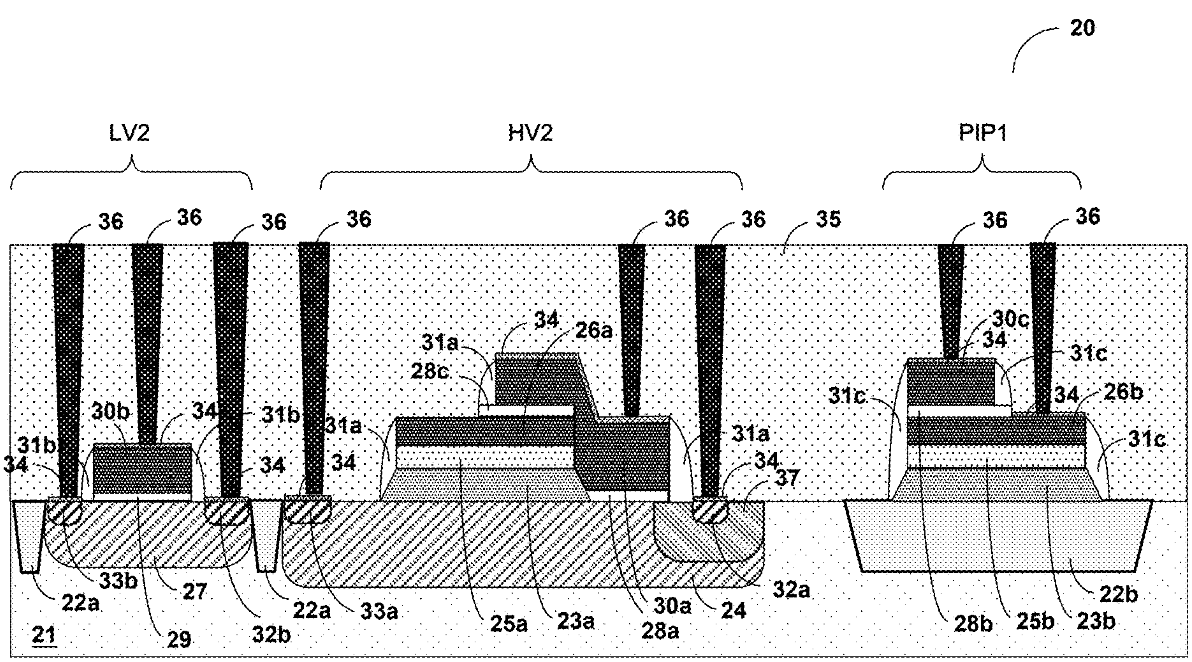
Publication Classification

(51) **Int. Cl.**
H01L 21/8234 (2006.01)
H01L 21/265 (2006.01)

(52) **U.S. Cl.**
CPC ***H10D 84/038*** (2025.01); ***H01L 21/2652*** (2013.01); ***H10D 1/68*** (2025.01); ***H10D 84/0142*** (2025.01); ***H10D 84/0144*** (2025.01); ***H10D 84/0156*** (2025.01)

(57) **ABSTRACT**

A manufacturing method of a semiconductor integrated structure having a high voltage device, a low voltage device and a capacitor, includes: forming a bottom thermal oxide layer on a substrate; forming a chemical vapor deposition (CVD) oxide layer; forming a poly silicon hard mask layer; etching the poly silicon hard mask layer to form a high voltage poly silicon hard mask and a first electrode plate simultaneously; etching the CVD oxide layer and using the high voltage poly silicon hard mask and the first electrode plate as etching barrier layers to form a high voltage CVD oxide region and a capacitor CVD oxide region simultaneously; etching the bottom thermal oxide layer and using the high voltage poly silicon hard mask and the first electrode plate as the etching barrier layers to form a high voltage bottom thermal oxide region and a bottom thermal oxide region simultaneously.



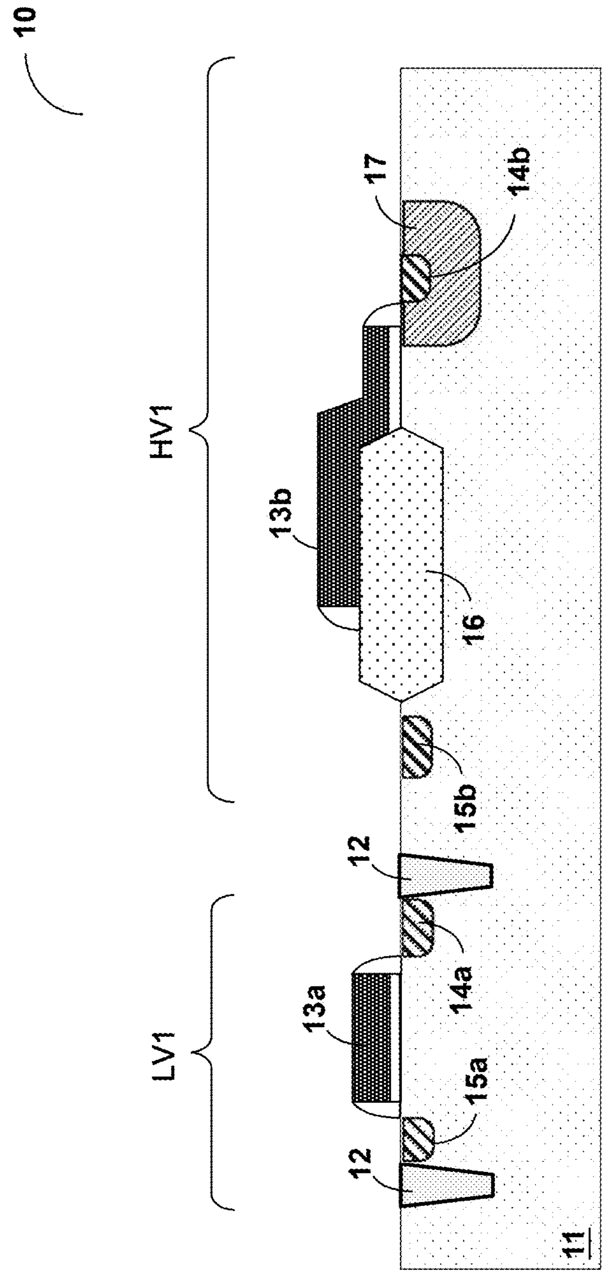


Fig. 1 (Prior Art)

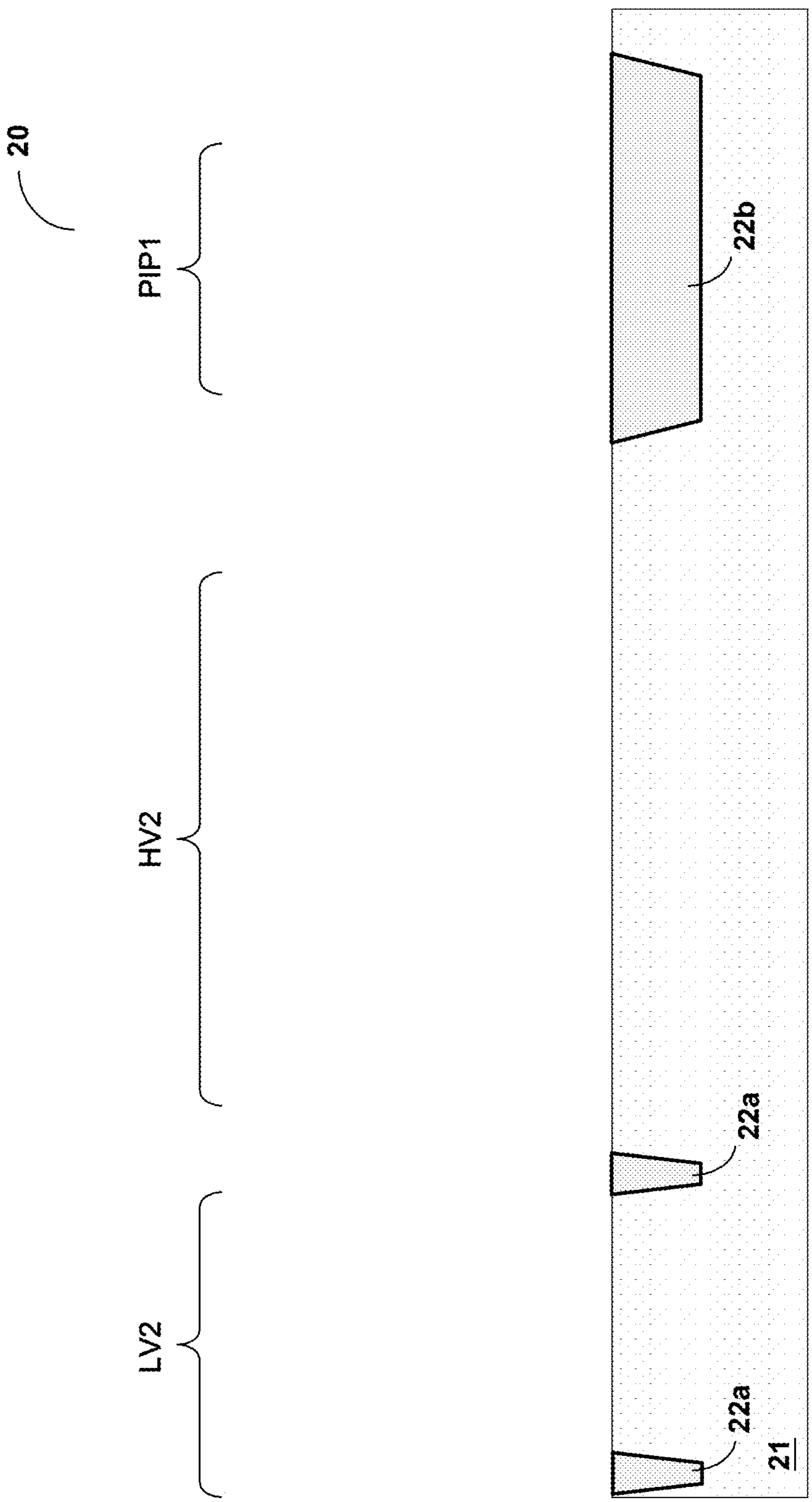


Fig. 2A

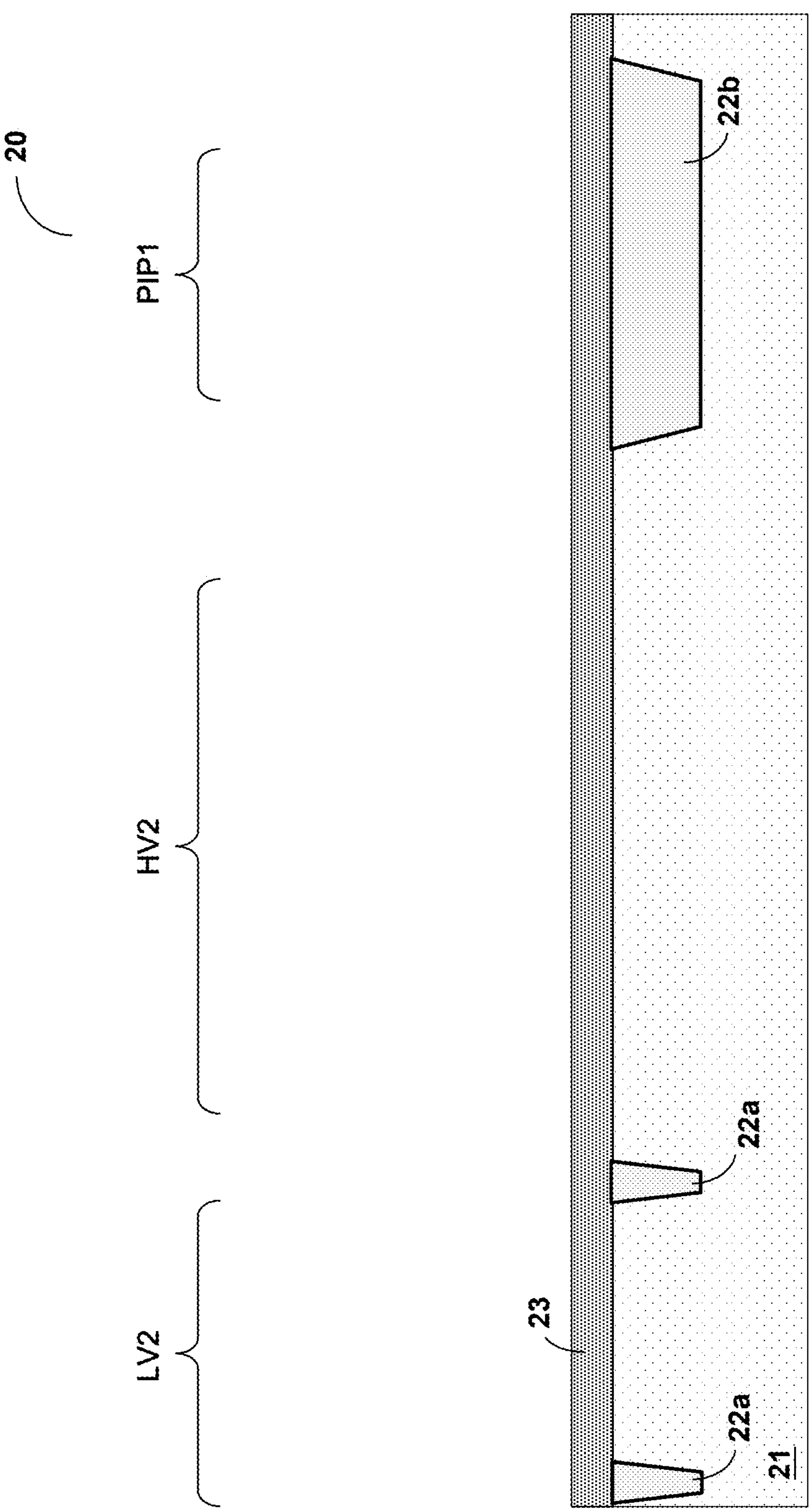


Fig. 2B

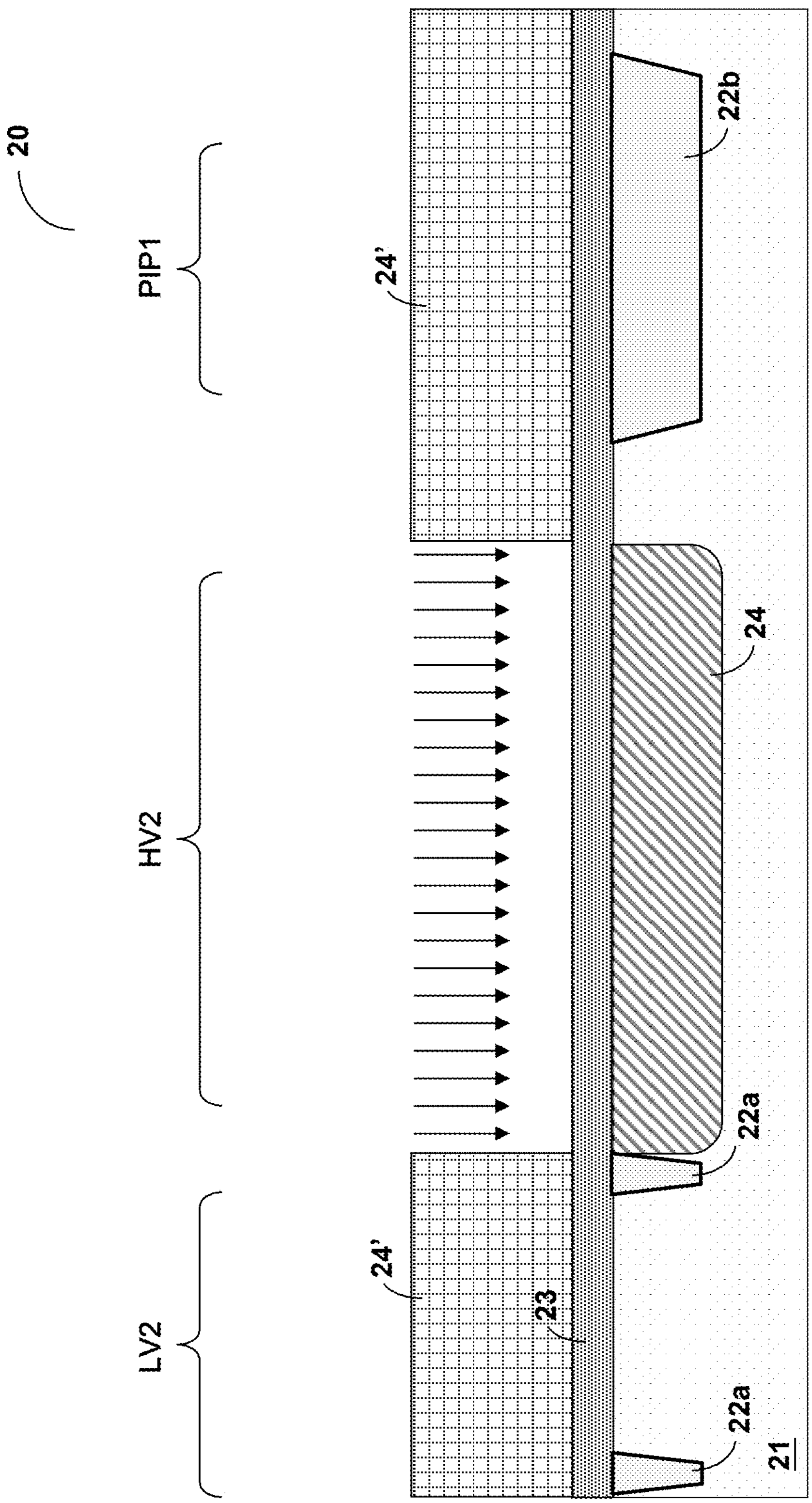


Fig. 2C

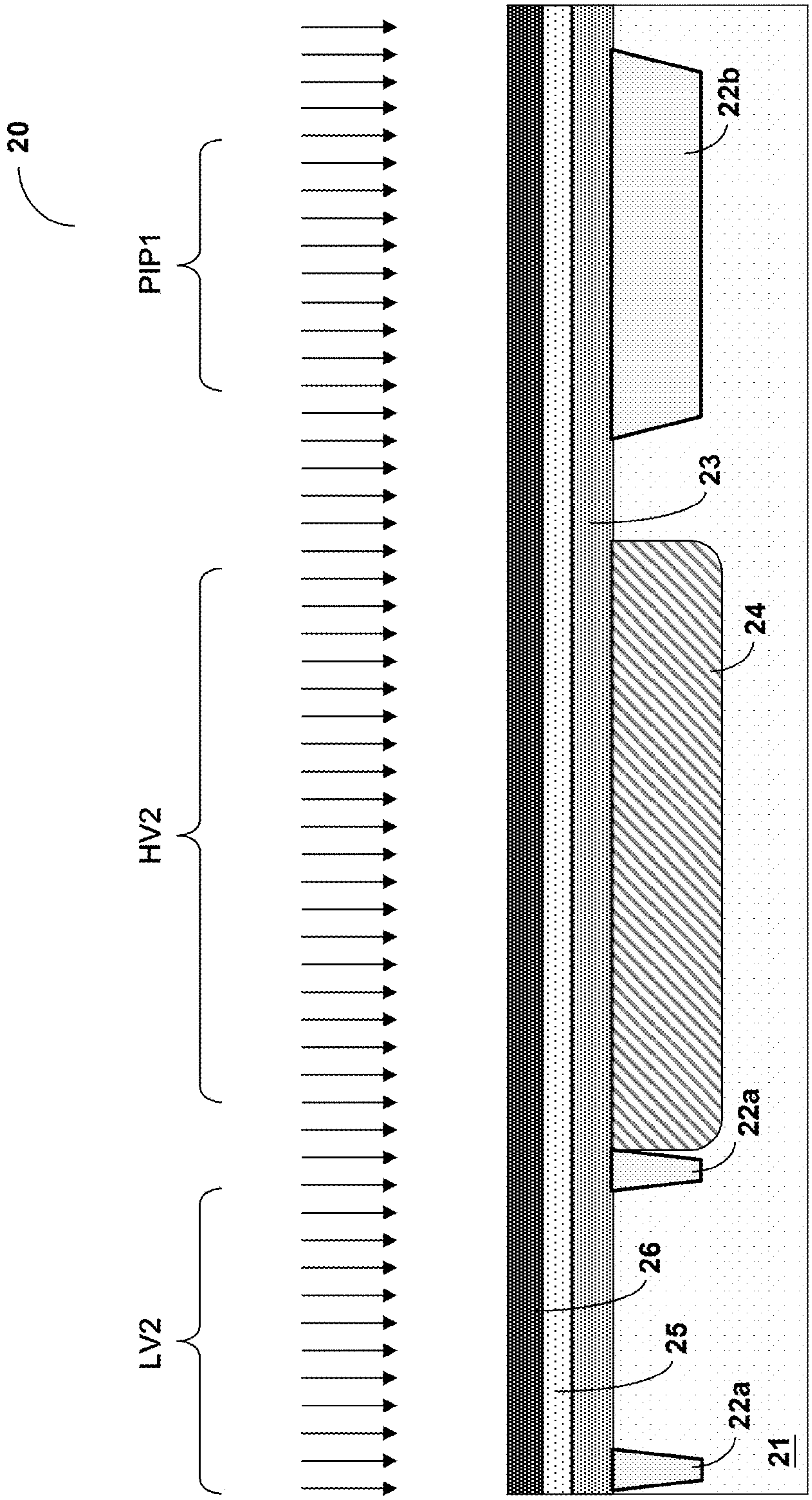


Fig. 2D

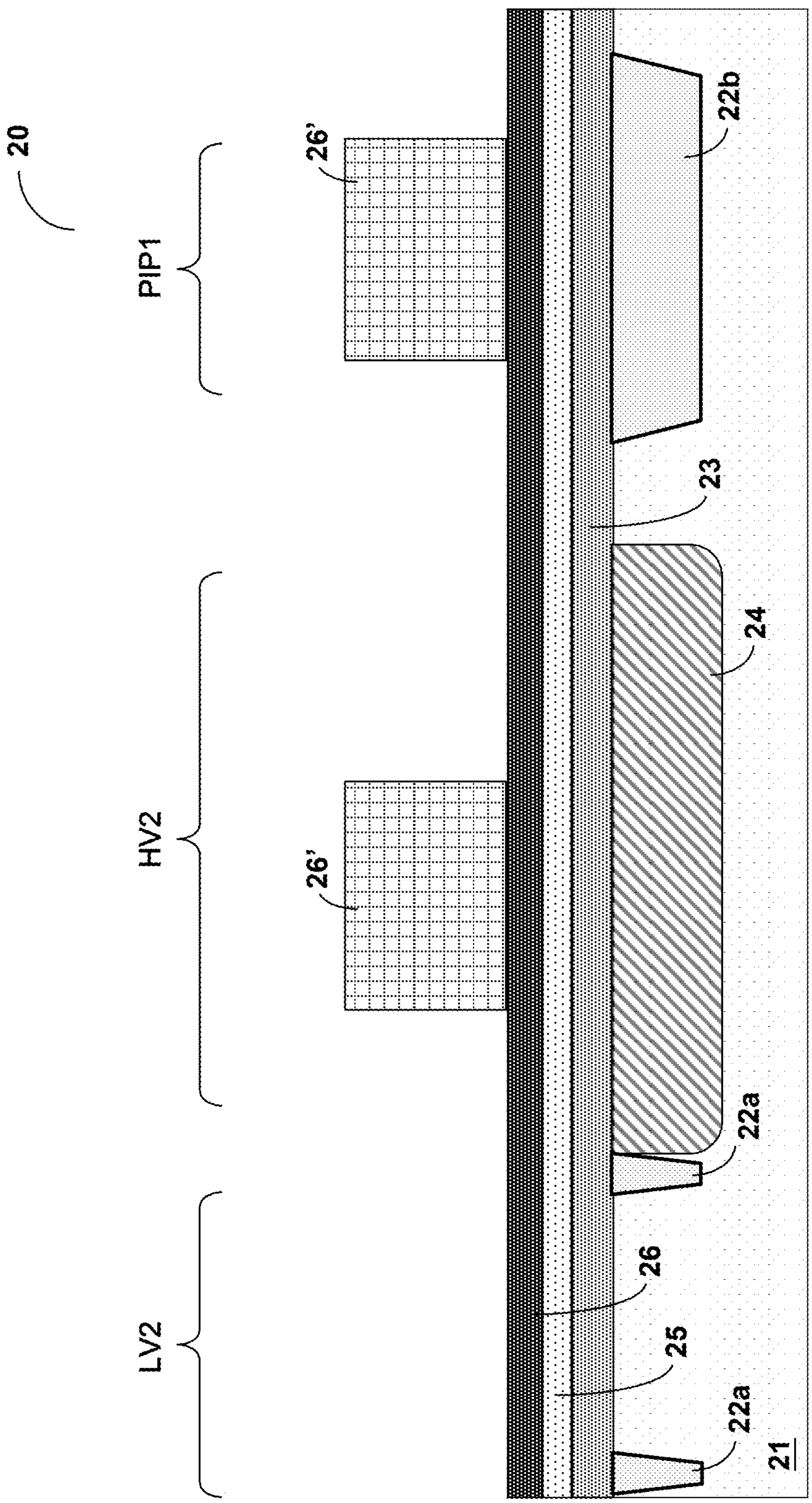


Fig. 2E

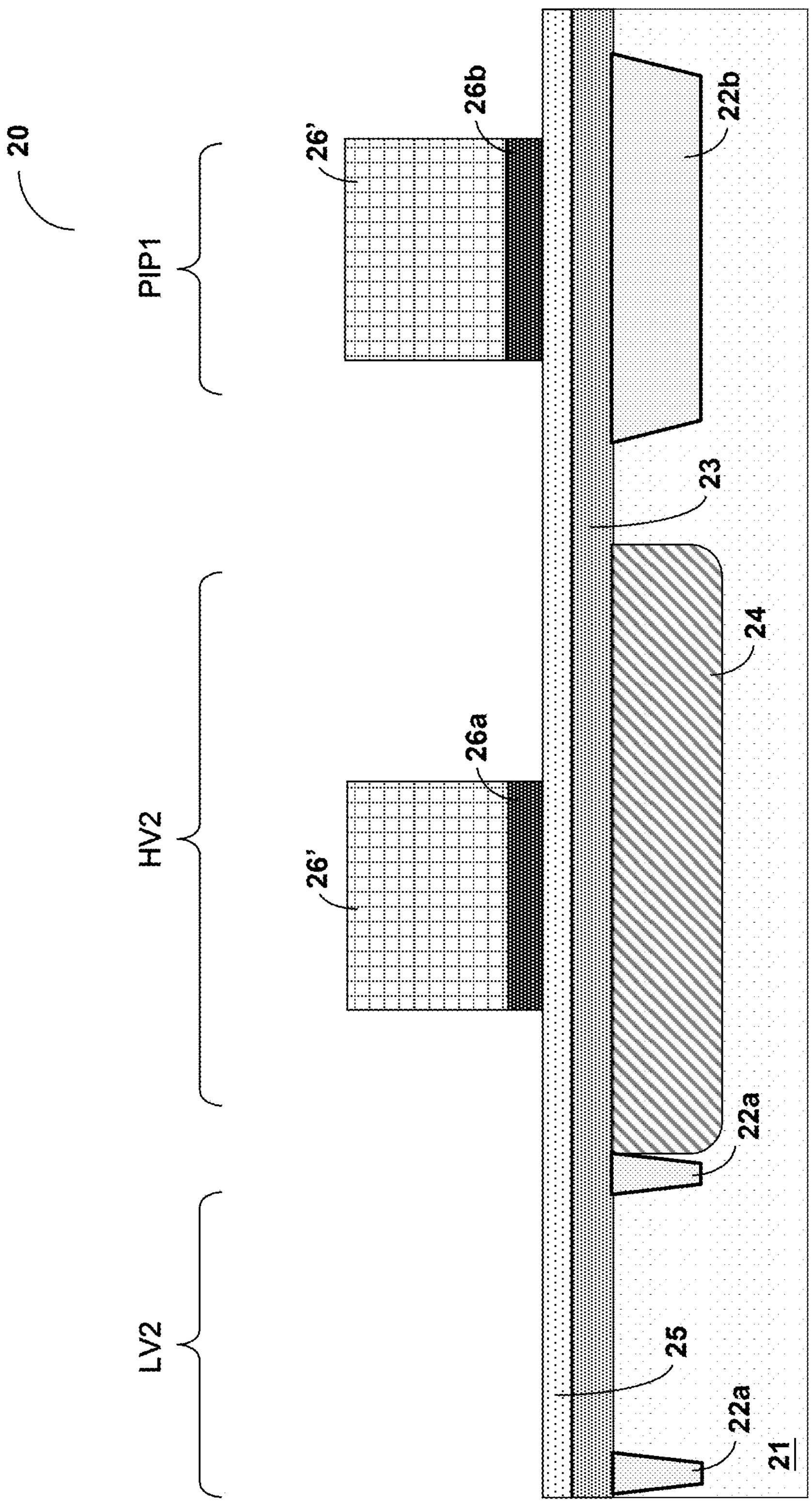


Fig. 2F

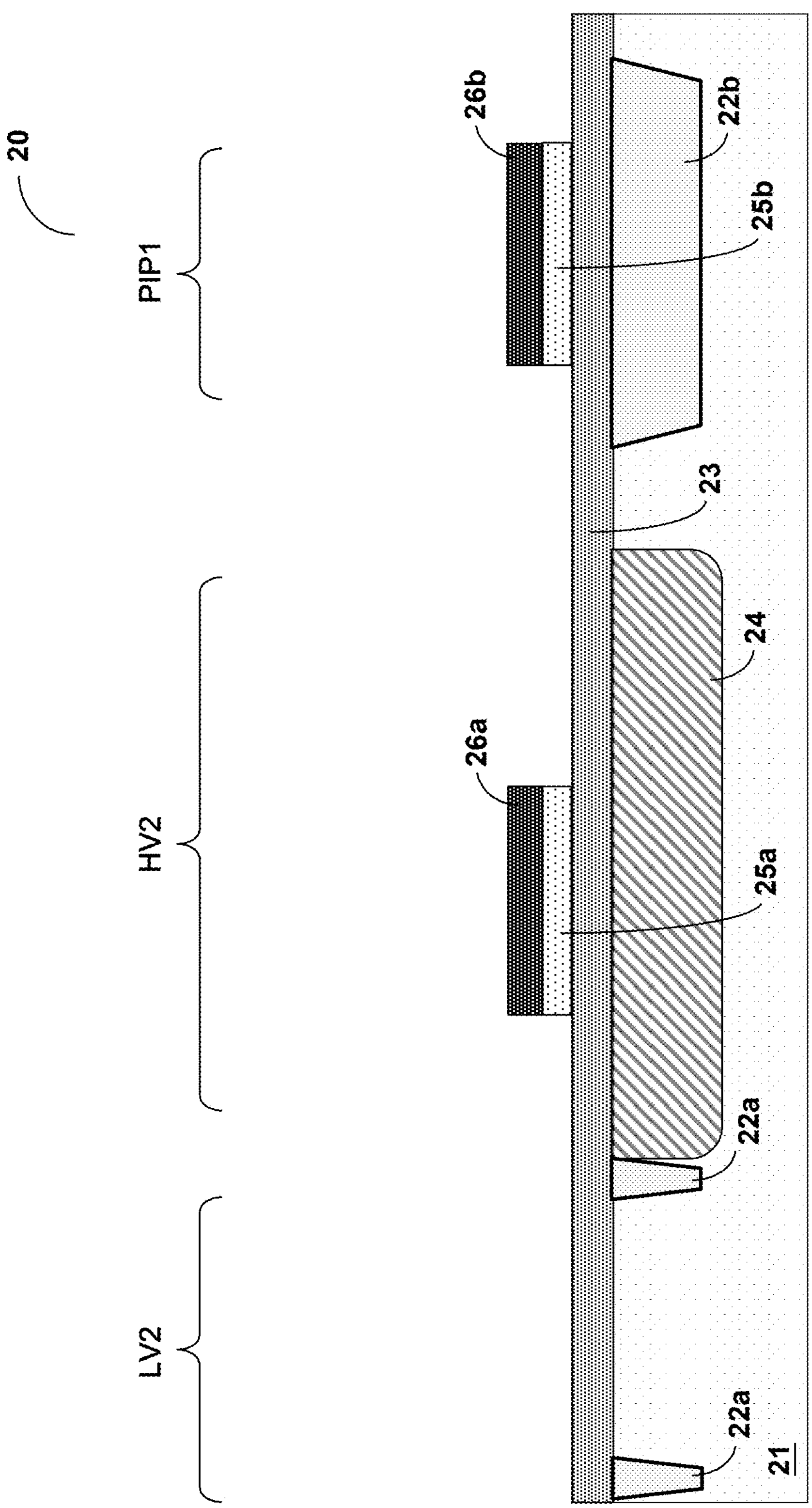


Fig. 2G

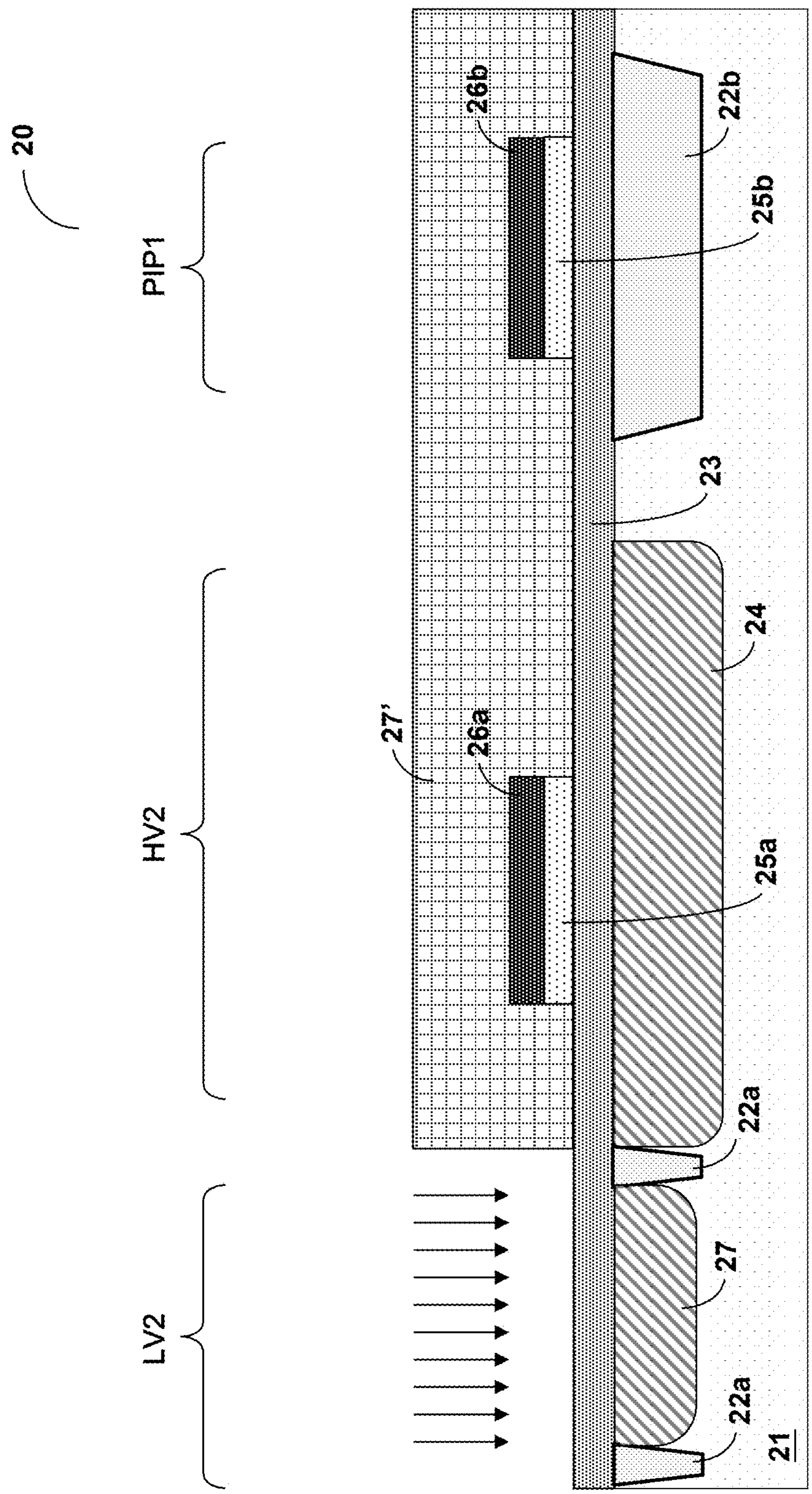


Fig. 2H

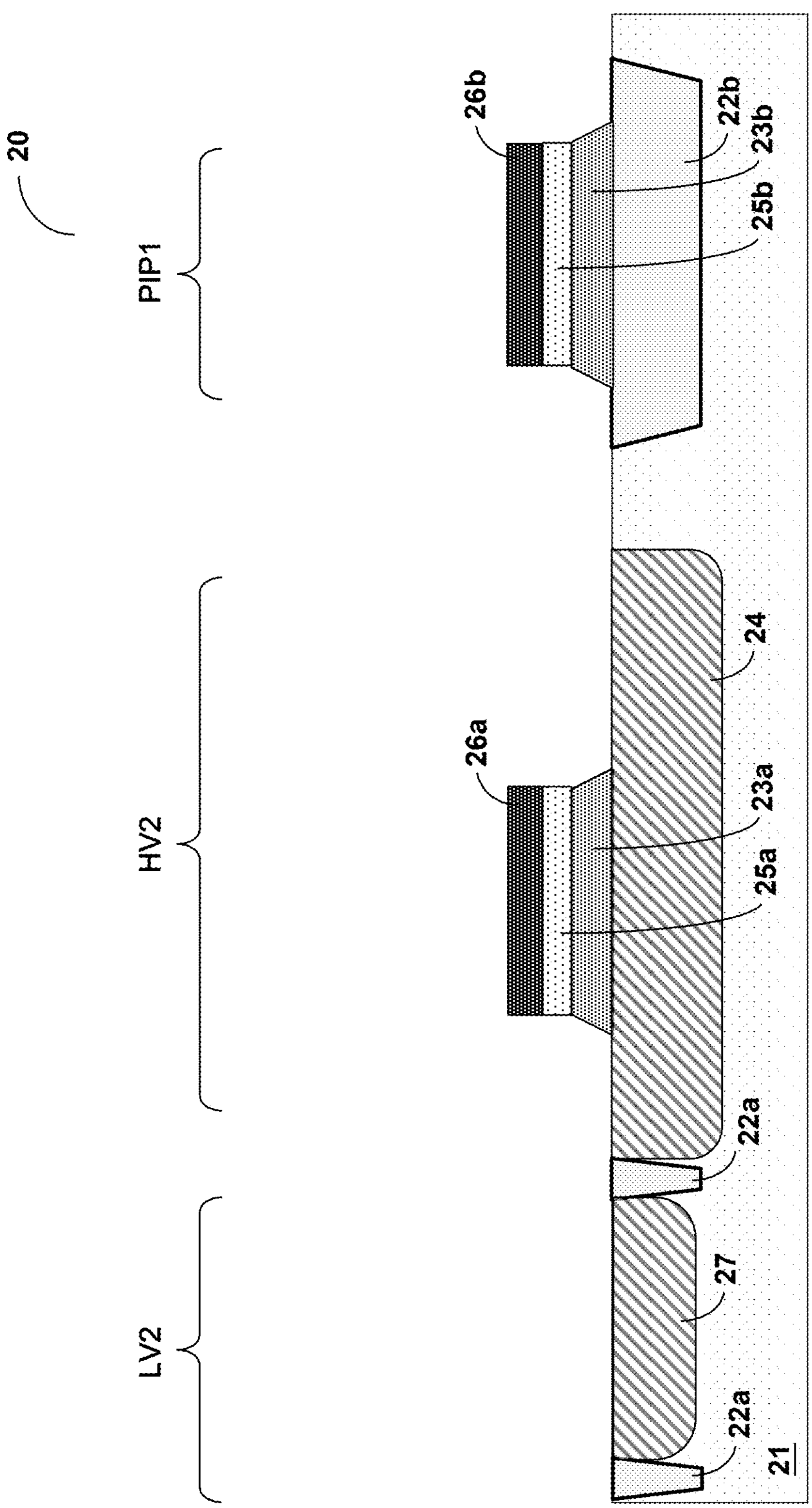


Fig. 2I

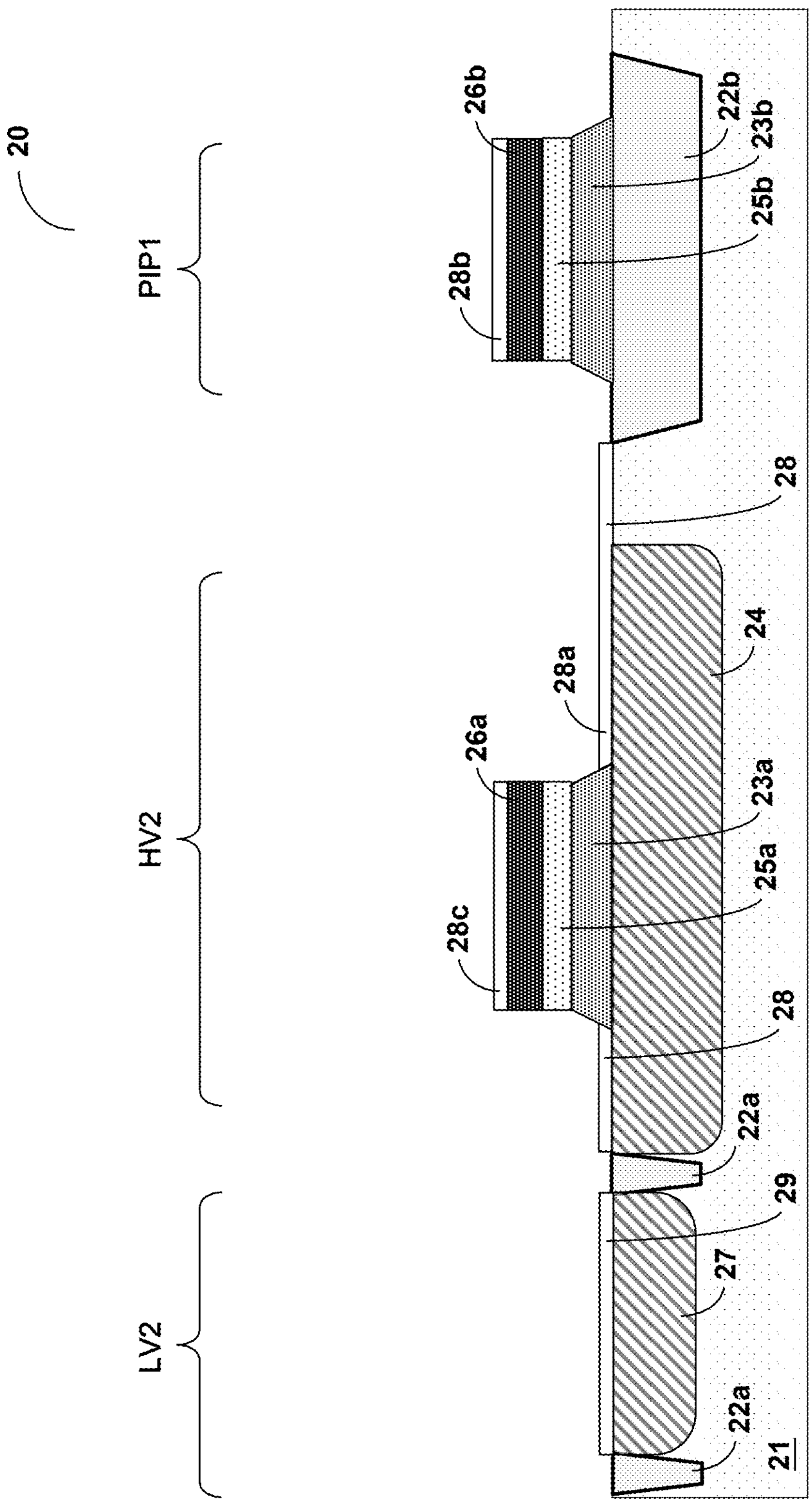

20
PIP1
HV2
LV2
26b
22b
23b
25b
28b
28
24
28a
26a
23a
25a
28c
28
22a
29
27
22a
21

Fig. 2J

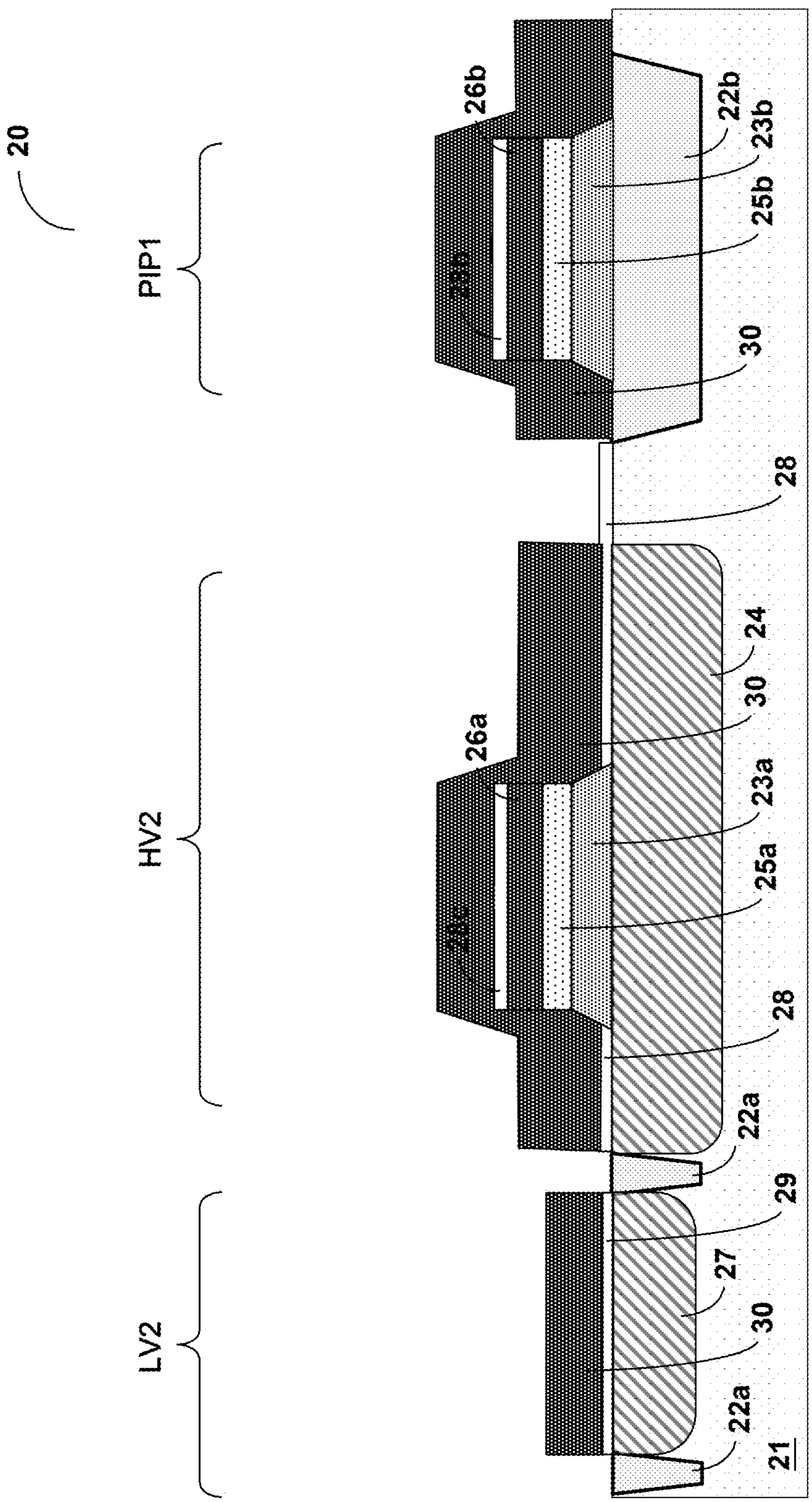

20
PIP1
HV2
LV2
26b
22b
23b
25b
28b
30
28
24
26a
23a
25a
28c
22a
29
27
21

Fig. 2K

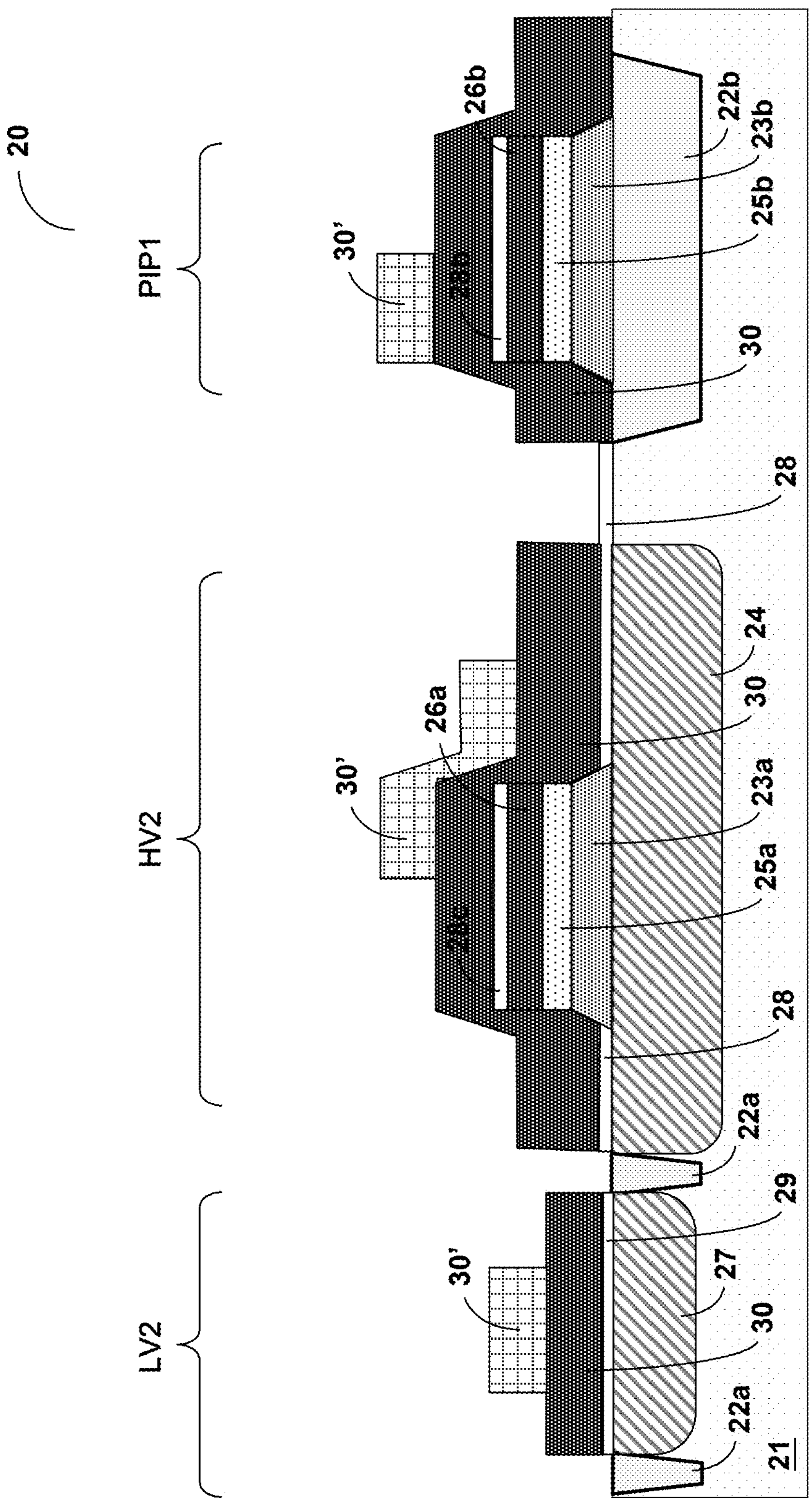

20
PIP1
HV2
LV2
30'
26b
22b
23b
25b
28b
30
28
24
26a
23a
25a
28c
22a
29
27
21

Fig. 2L

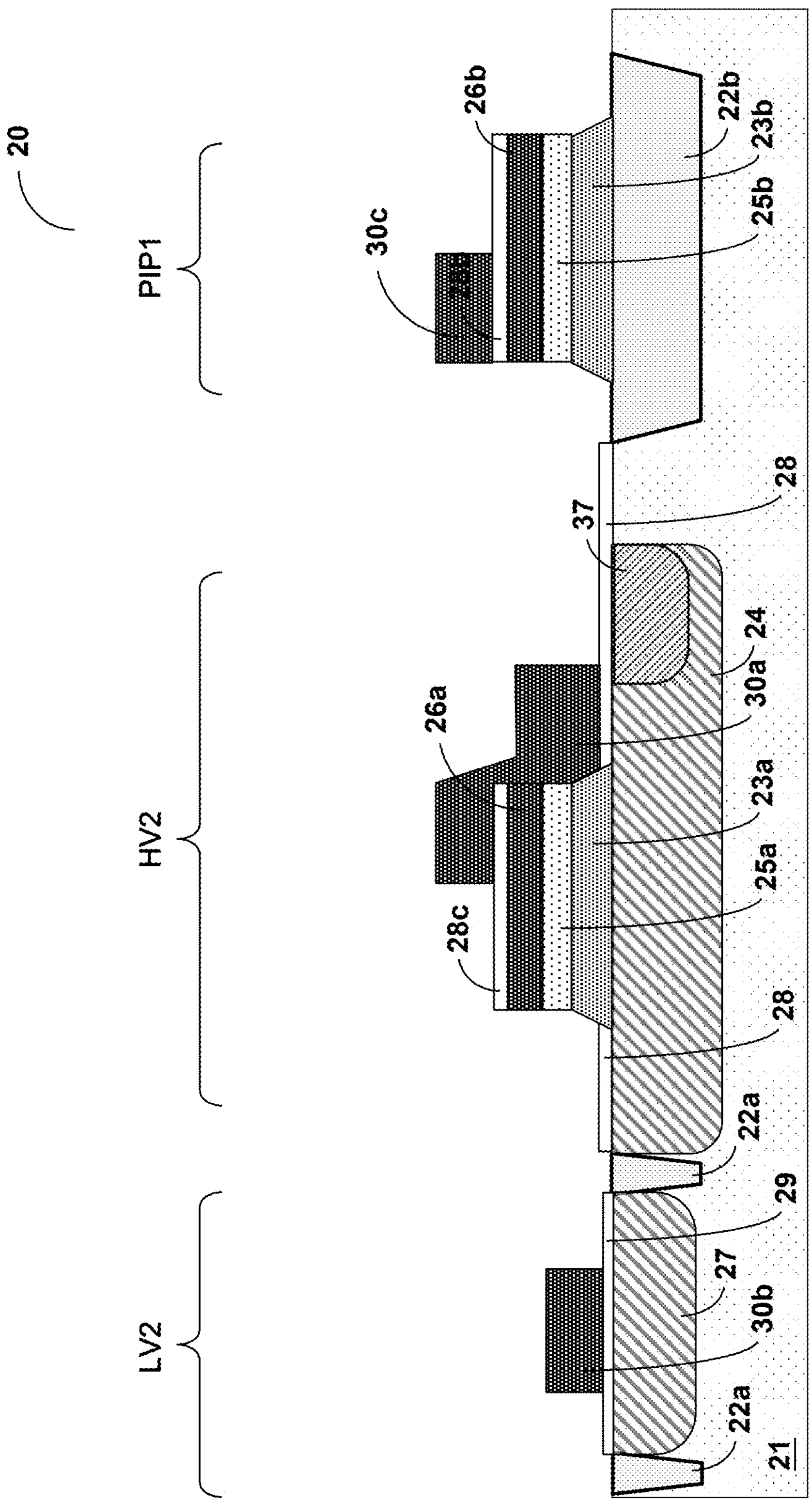

20
PIP1
HV2
LV2
26b
22b
23b
25b
30c
28c
28
37
24
30a
26a
23a
25a
28c
28
22a
29
27
30b
22a
21

Fig. 2M

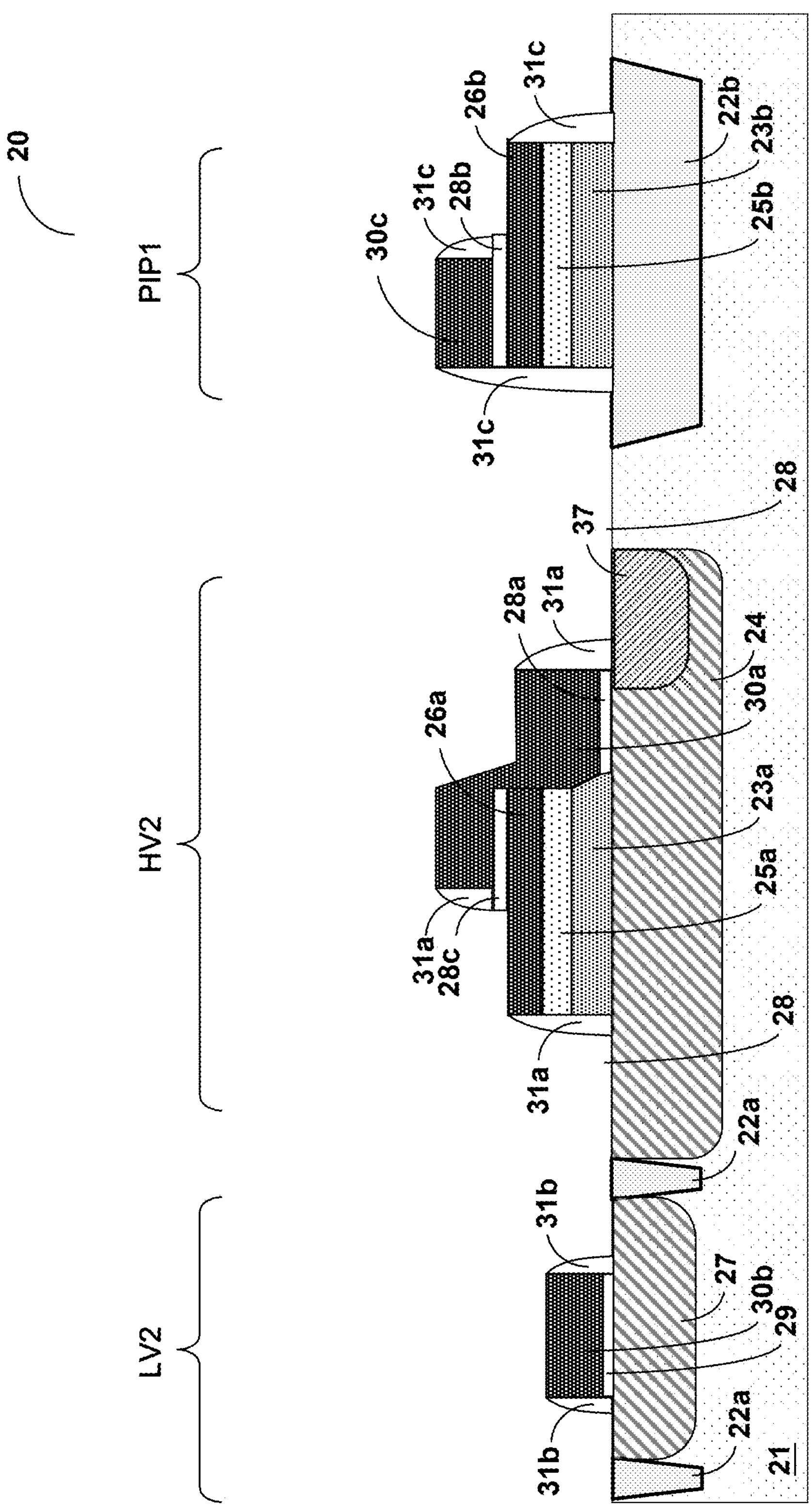

20
PIP1
HV2
LV2
31c
26b
22b
23b
25b
31c
28b
30c
31c
28
37
28a
31a
24
30a
26a
23a
25a
31a
28c
28
31a
22a
31b
27
30b
29
22a
31b
21

Fig. 2N

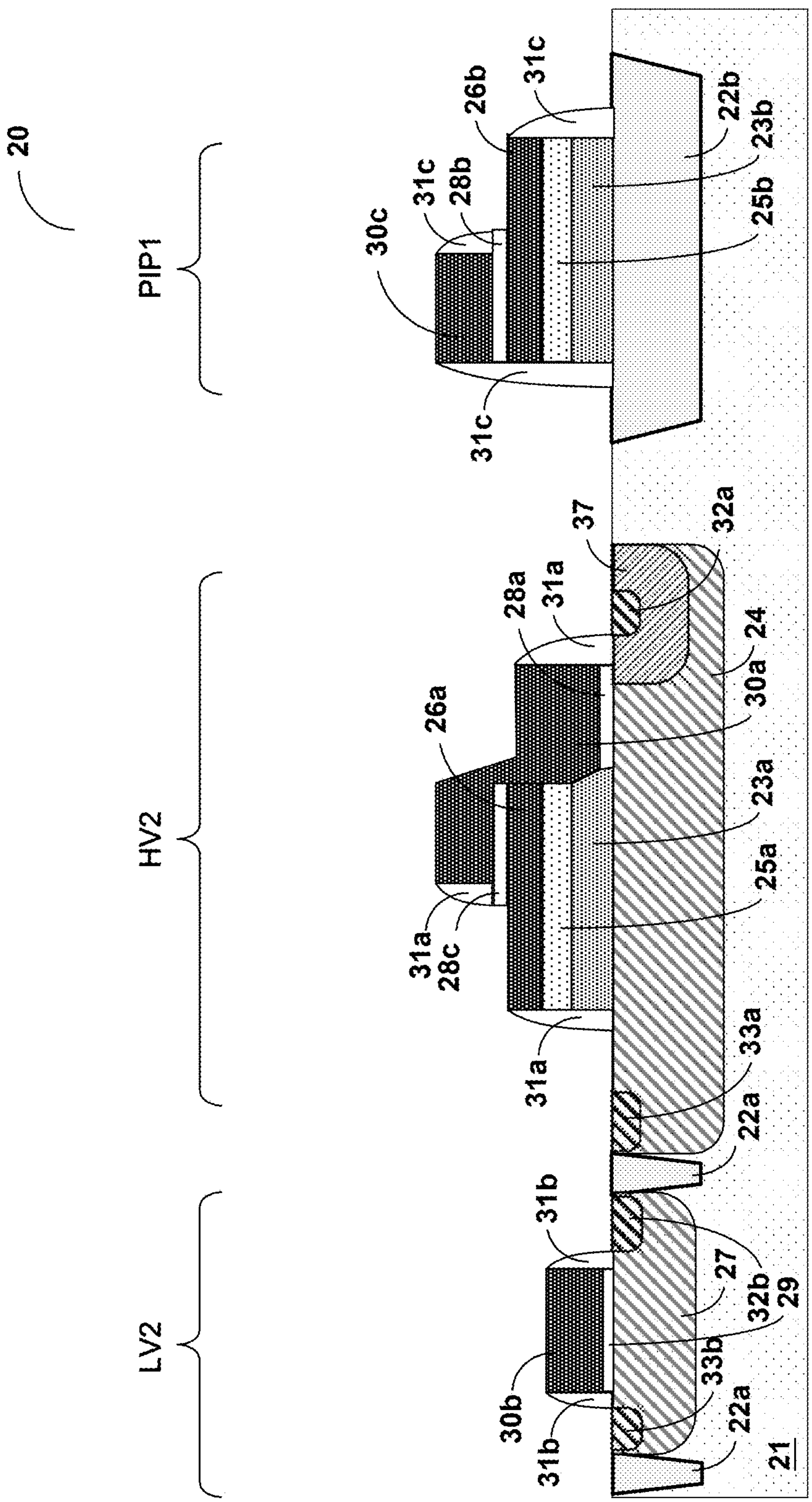

20
PIP1
HV2
LV2
30c
31c
28b
26b
31c
22b
23b
25b
31c
37
32a
31a
28a
24
30a
26a
23a
25a
31a
28c
31a
33a
22a
31b
27
32b
29
33b
30b
31b
22a
21

Fig. 2O

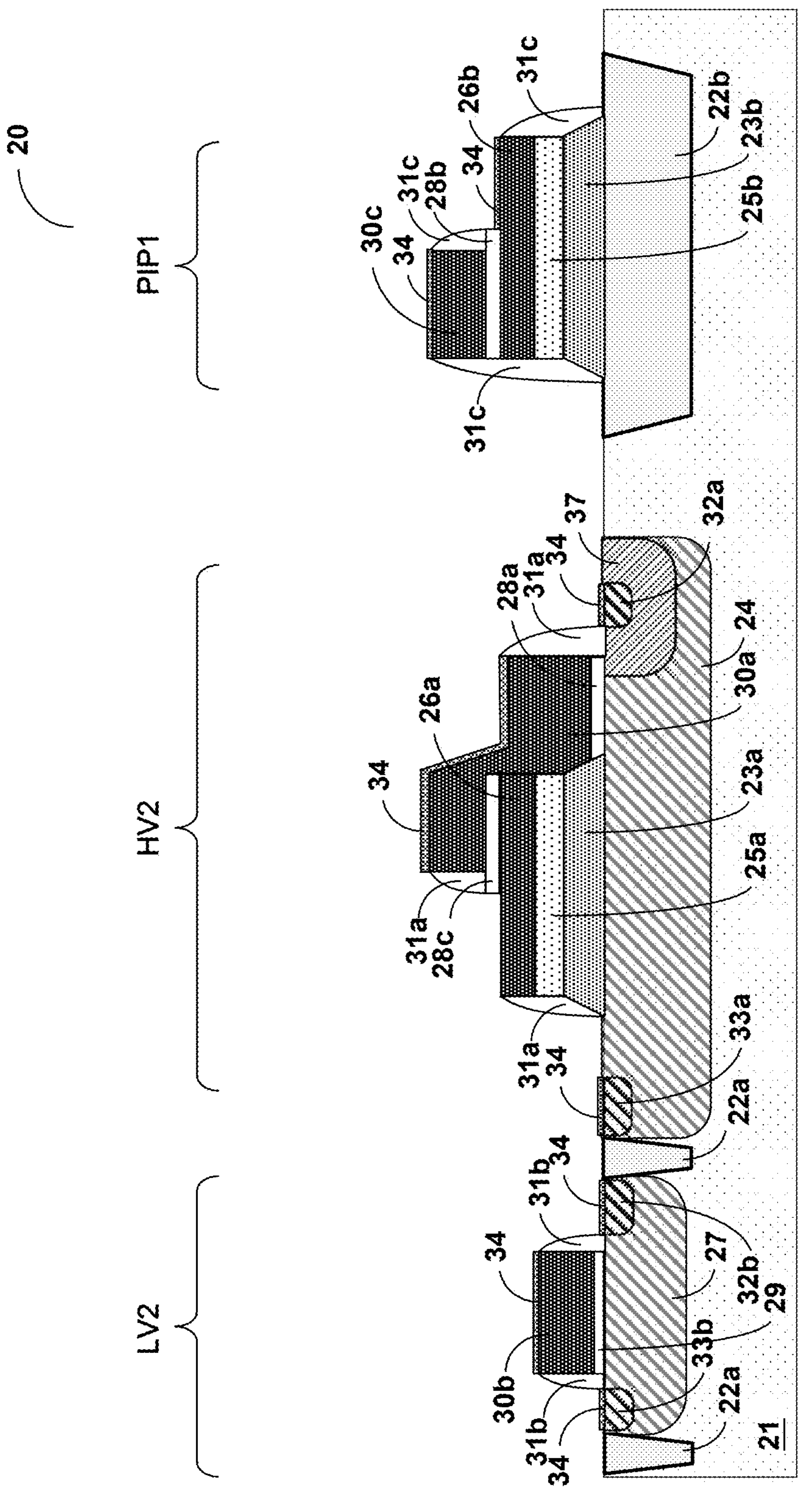

20
PIP1
HV2
LV2
26b
31c
22b
23b
25b
34
31c
28b
30c
37
32a
34
31a
28a
24
30a
26a
23a
25a
31a
28c
31a
33a
22a
31b
27
32b
29
33b
30b
22a
31b
21

Fig. 2P

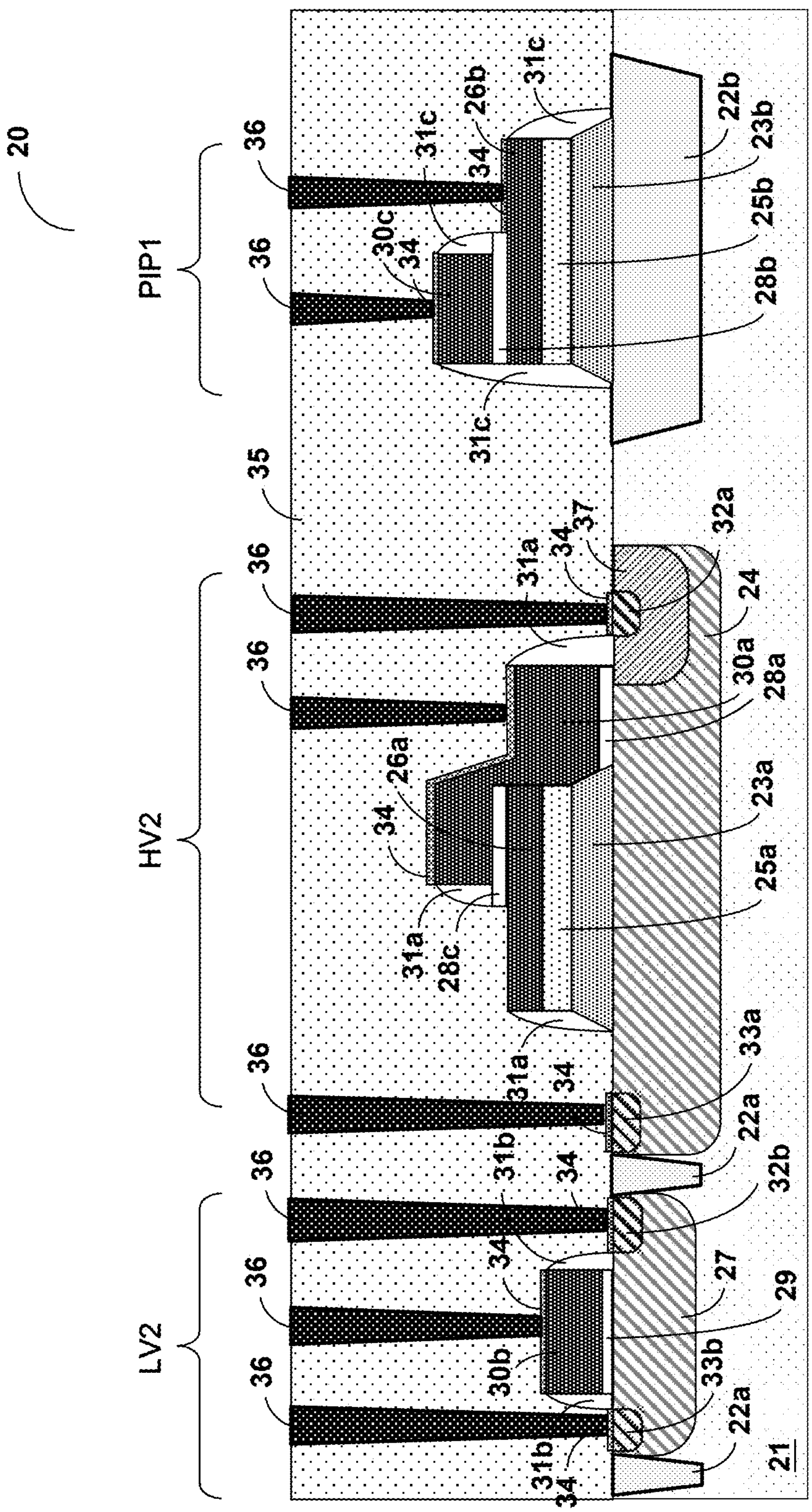

20
PIP1
HV2
LV2
36
34
35
37
31a
31b
31c
30a
30b
30c
28a
28b
28c
26a
26b
25a
25b
23a
23b
22a
22b
24
27
29
32a
32b
33a
33b
21

Fig. 2Q

MANUFACTURING METHOD OF SEMICONDUCTOR INTEGRATED STRUCTURE HAVING HIGH/LOW VOLTAGE DEVICES AND CAPACITOR

CROSS REFERENCE

[0001] The present invention claims priority to TW 113108618 filed on Mar. 8, 2024.

BACKGROUND OF THE INVENTION

Field of Invention

[0002] The present invention relates to a manufacturing method of a semiconductor integrated structure having high and low voltage devices and a capacitor, particularly, it relates to such semiconductor integrated structure which integrates a high voltage device, a low voltage device, and a poly silicon-insulator-poly silicon (PIP) capacitor.

Description of Related Art

[0003] Referring to FIG. **1**, which shows a sectional schematic view of a prior art semiconductor integrated structure **10** comprising a high device HV**1**, a low voltage device LV**1** and a capacitor (not shown). As illustrated in FIG. **1**, the semiconductor integrated structure **10** is formed on a substrate **11**, comprising plural insulation regions **12**, a low voltage gate **13***a*, a high voltage gate **13***b*, a low voltage source **14***a*, a high voltage source **14***b*, a low voltage drain **15***a*, a high voltage drain **15***b*, a reduce surface electric field (RESURF) oxide region **16**, and a body region **17**.

[0004] The plural insulation regions **12** serve to electrically isolate the low voltage device LV**1** from the high voltage device HV**1** in the substrate **11**. The low voltage gate **13***a* is formed on the substrate **11** of the low voltage device LV**1**, with the low voltage source **14***a* and the low voltage drain **15***a* also formed in the substrate **11**. Meanwhile, the high voltage gate **13***b* and the RESURF oxide region **16** are formed on the substrate **11** of the high voltage device HV**1**, with the high voltage source **14***b* and the high voltage drain **15***b* also formed in the same substrate **11**.

[0005] In a typical LDMOS device, as the high voltage device HV**1** shown in FIG. **1**, employing a field plate structure (the RESURF oxide region **16** and an extension of the high voltage gate **13***b*) is the most effective method to improve device breakdown voltage and to mitigate issues of hot carrier injection near the gate edge. This approach alleviates localized electric field crowding that could lead to early device failure. Although a traditional LOCOS field oxide region, as illustrated for the RESURF oxide region **16** in FIG. **1**, is the simplest way to implement a field plate structure without additional mask definitions, they can cause severe issues such as increased on-resistance during high-temperature annealing processes or boron diffusion outward from LDMOS devices.

[0006] To resolve these issues, a field plate composed of a dielectric layer with an angled profile relative to the surface of the substrate **11**, typically made of silicon dioxide, but also potentially of silicon nitride, silicon oxynitride, or high-k materials, is used between the high voltage gate **13***b* and the substrate **11** to implement the RESURF technique.

[0007] However, this angled dielectric layer structure might involve additional oxide growth thermal processes that lead to exceeding the thermal budget; furthermore, defining this angled dielectric layer can result in shallow trench isolation (STI) corner loss, also known as notching, due to additional wet etching processes. Consequently, prior art has introduced a novel two-step angled dielectric layer oxidation process and a maskless poly silicon-insulator-poly silicon (PIP) capacitor structure to overcome these challenges.

[0008] Additionally, the prior art encounters challenges with photoresist stripping during wet etching steps, which can further damage the shallow trench isolation or LOCOS field oxide regions; additionally, multiple thermal oxidation processes also tend to exceed the thermal budget constraints.

[0009] In view of the above, the present invention proposes a manufacturing method of a semiconductor integrated structure having high and low voltage devices and a capacitor, which avoids damage to the shallow trench isolation or LOCOS field oxide regions and reduces the number of thermal oxidation processes, thus addressing problems related to notching and exceeding thermal budget limitations.

SUMMARY OF THE INVENTION

[0010] From one perspective, the present invention provides a manufacturing method of integrated an structure of a semiconductor integrated structure having high and low voltage devices and a capacitor, comprising: forming a bottom thermal oxide layer on a substrate, wherein the bottom thermal oxide layer completely covers a high voltage device area, a low voltage device area, and a capacitor area of the substrate; forming a high voltage well in the substrate of the high voltage device area; forming a chemical vapor deposition (CVD) oxide layer that completely covers the bottom thermal oxide layer; forming a polysilicon hard mask layer that completely covers the CVD oxide layer; etching the polysilicon hard mask layer to simultaneously form a high voltage polysilicon hard mask in the high voltage device area and a first electrode plate in the capacitor area; etching the CVD oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form a high voltage CVD oxide region in the high voltage device area and a capacitor CVD oxide region in the capacitor area; forming a low voltage well in the substrate of the low voltage device area; etching the bottom thermal oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form a high voltage bottom thermal oxide region in the high voltage device area and a bottom thermal oxide region in the capacitor area; forming a high voltage gate oxide layer over the high voltage device area on the substrate; forming a low voltage gate oxide layer over the low voltage device area on the substrate; when forming the high voltage gate oxide layer or the low voltage gate oxide layer, simultaneously forming a capacitor dielectric layer, connected and fully covering the first electrode plate; forming a gate polysilicon layer, connected and fully covering the high voltage gate oxide layer, the low voltage gate oxide layer, and the capacitor dielectric layer; and etching the gate polysilicon layer to simultaneously form a high voltage gate in the high voltage device area, a low voltage gate in the low voltage device area, and a second electrode plate in the capacitor area.

[0011] In one embodiment, the step of forming the low voltage well in the substrate of the low voltage device area includes: using an ion implantation process step, with the bottom thermal oxide layer serving as a sacrificial layer, and accelerating ions to penetrate the sacrificial layer to implant into the low voltage device area to form the low voltage well.

[0012] In one embodiment, the step of etching the CVD oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form the high voltage CVD oxide region in the high voltage device area and the capacitor CVD oxide region in the capacitor area includes: using a wet etching process step to etch the CVD oxide layer.

[0013] In one embodiment, the step of etching the bottom thermal oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form the high voltage bottom thermal oxide region in the high voltage device area and the bottom thermal oxide region in the capacitor area includes: using a wet etching process step to etch the bottom thermal oxide layer, such that the high voltage side wall of the high voltage bottom thermal oxide region and the capacitor side wall of the bottom thermal oxide region each have an inclined angle with respect to the upper surface of the substrate.

[0014] In one embodiment, the manufacturing method further includes: after forming the polysilicon hard mask layer, using an ion implantation process step to accelerate N-type or P-type ions to implant into the polysilicon hard mask layer.

[0015] In one embodiment, the manufacturing method further includes: after forming the gate polysilicon layer, using an ion implantation process step to accelerate N-type or P-type ions to implant into the gate polysilicon layer.

[0016] In one embodiment, the high voltage gate is in direct contact with the high voltage polysilicon hard mask.

[0017] In one embodiment, the manufacturing method further includes: after forming the high voltage gate, the low voltage gate, and the second electrode plate, forming two high voltage spacers corresponding to and connected on either side of the high voltage gate, and two low voltage spacers corresponding to and connected on either side of the low voltage gate.

[0018] In one embodiment, the manufacturing method further includes: after forming the high voltage gate, the low voltage gate, and the second electrode plate, forming two capacitor spacers corresponding to and connected on either side of the second electrode plate.

[0019] In one embodiment, the manufacturing method further includes: simultaneously forming a high voltage source and a high voltage drain in the high voltage device area, and a low voltage source and a low voltage drain in the low voltage device area.

[0020] In one embodiment, the manufacturing method further includes: using a silicidation metal process step to simultaneously form a plurality of silicide metal layers corresponding to the upper surfaces of the high voltage gate, the high voltage source, the high voltage drain, the low voltage gate, the low voltage source, the low voltage drain, the first electrode plate, and the second electrode plate.

[0021] In one embodiment, the manufacturing method further includes: forming an inter-layer dielectric (ILD) layer on the substrate, completely covering the high voltage gate, the low voltage gate, and the second electrode plate.

[0022] In one embodiment, the manufacturing method further includes: forming a plurality of electrical contact plugs in the ILD layer, electrically connecting the multiple silicide metal layers.

[0023] In one embodiment, the manufacturing method further includes: forming a high voltage body region in the high voltage well of the high voltage device area, wherein the high voltage source is located in the high voltage body region.

[0024] The objectives, technical details, features, and effects of the present invention will be better understood with regard to the detailed description of the embodiments below, with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** is a schematic cross-sectional view illustrating an semiconductor integrated structure **10** comprising a high device HV**1**, a low voltage device LV**1** and a capacitor.

[0026] FIGS. **2**A-**2**Q are schematic cross-sectional views illustrating a manufacturing method of a semiconductor integrated structure **20** having high and low voltage devices and a capacitor according to an embodiment of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0027] The drawings as referred to throughout the description of the present invention are for illustration only, to show the interrelations among the process steps and the layers, while the shapes, thicknesses, and widths are not drawn in actual scale.

[0028] Referring to FIGS. **2**A-**2**Q, according to an embodiment of the present invention, a cross-sectional schematic view of a method for manufacturing a semiconductor integrated structure **20** having high and low voltage devices and a capacitor is shown. As illustrated in FIG. **2**A, a substrate **21** is first provided, and insulation regions **22***a* and **22***b* are formed on the substrate **21**. The substrate **21** may be, for example but not limited to, a P-type or N-type semiconductor substrate. This is well known to those of ordinary skill in the art and is not further elaborated here. As shown in FIG. **2**B, a bottom thermal oxide layer **23** is then formed on substrate **21**, where the bottom thermal oxide layer **23** completely covers a high voltage device area HV**2**, a low voltage device area LV**2**, and a capacitor area PIP**1**. Subsequently, as shown in FIG. **2**C, a high voltage well area **24** is formed in the high voltage device area HV**2**. The high voltage well area **24** is formed, for example, by an ion implantation process step, where N-type or P-type impurities are implanted into the area defined by a photomask **24'** in the form of accelerated ions, which is also well known in the art and not elaborated here. Next, as shown in FIG. **2**D, a chemical vapor deposition (CVD) oxide layer **25** is formed completely covering the bottom thermal oxide layer **23**. As shown in FIG. **2**D, a polysilicon hard mask layer **26** is then formed completely covering the CVD oxide layer **25**. In one embodiment, after forming the polysilicon hard mask layer **26**, N-type or P-type ions are accelerated and implanted into the polysilicon hard mask layer **26**, for example, by an ion implantation process step.

[0029] Continuing, as shown in FIGS. **2**E and **2**F, the polysilicon hard mask layer **26** is etched using a photomask

26' to simultaneously form a high voltage polysilicon hard mask **26***a* in the high voltage device area HV**2** and a first electrode plate **26***b* in the capacitor area. Then, as shown in FIG. **2**G, for example, a wet etching process step is used to etch the CVD oxide layer **25**, utilizing the high voltage polysilicon hard mask **26***a* and the first electrode plate **26***b* as etching barrier layers to simultaneously form a high voltage CVD oxide region **25***a* in the high voltage device area HV**2** and a capacitor CVD oxide region **25***b* in the capacitor area PIP**1**. The formation of the high voltage CVD oxide region **25***a*, for example, may be performed by a deposition process step, which is also well known in the art and not elaborated here.

[0030] Continuing, as shown in FIG. **2**H, a low voltage well area **27** is formed in the low voltage device area LV**2**. The low voltage well area **27** is formed, for example, by an ion implantation process step, where N-type or P-type impurities penetrate the sacrificial layer formed by the bottom thermal oxide layer **23** and are implanted into the low voltage device area LV**2** defined by a photomask **27**', using accelerated ions. Next, as shown in FIG. **2**I, the bottom thermal oxide layer **23** is etched using the high voltage polysilicon hard mask **26***a* and the first electrode plate **26***b* as etching barriers, to simultaneously form a high voltage bottom thermal oxide region **23***a* in the high voltage device area HV**2** and a capacitor bottom thermal oxide region **23***b* in the capacitor area PIP**1**. In one embodiment, for example, a wet etching process step is used to etch the bottom thermal oxide layer **23**, such that the high voltage sidewall of the high voltage bottom thermal oxide region **23***a* and the capacitor sidewall of the capacitor bottom thermal oxide region **23***b* each have an inclined angle with respect to the upper surface of substrate **21**.

[0031] Afterwards, as shown in FIG. **2**J, a high voltage gate oxide layer **28***c*, **28***a* is formed on the substrate **21** in the high voltage device area HV**2**, and a low voltage gate oxide layer **29** is formed on substrate **21** in the low voltage device area LV**2**. As shown in FIG. **2**J, when forming the high voltage gate oxide layer **28***c*, **28***a* or the low voltage gate oxide layer **29**, a capacitor dielectric layer **28***b* is simultaneously formed, connected and fully covering the first electrode plate **26***b*. Subsequently, as shown in FIG. **2**K, a gate polysilicon layer **30** is formed, connected and fully covering the high voltage gate oxide layers **28***c*, **28***a*, the low voltage gate oxide layer **29**, and the capacitor dielectric layer **28***b*. In one embodiment, after forming the gate polysilicon layer **30**, N-type or P-type ions are accelerated and implanted into the gate polysilicon layer **30**, for example, by an ion implantation process step. Subsequently, as shown in FIGS. **2**L and **2**M, the gate polysilicon layer **30** is etched using a photomask **30'** to simultaneously form a high voltage gate **30***a* in the high voltage device area HV**2**, a low voltage gate **30***b* in the low voltage device area LV**2**, and a second electrode plate **30***c* in the capacitor area PIP**1**. As shown in FIG. **2**M, the high voltage gate **30***a* is in direct contact with the high voltage polysilicon hard mask **26***a*. As shown in FIG. **2**M, a high voltage body region **37** is formed in the high voltage well area **24** of the high voltage device area HV**2**, and the high voltage source (described later) is located within the high voltage body region **37**.

[0032] As shown in FIG. **2**N, subsequently, after forming the high voltage gate **30***a*, the low voltage gate **30***b*, and the second electrode plate **30***c*, two high voltage spacers **31***a* are formed corresponding and connected on either side of the high voltage gate **30***a*, and two low voltage spacers **31***b* are formed corresponding and connected on either side of the low voltage gate **30***b*. As shown in FIG. **2**N, in one embodiment, after forming the high voltage gate **30***a*, the low voltage gate **30***b*, and the second electrode plate **30***c*, two capacitor spacers **31***c* are formed corresponding and connected on either side of the second electrode plate **30***c*. As shown in FIG. **2**O, subsequently, a high voltage source **32***a* and a high voltage drain **33***a* are simultaneously formed in the high voltage device area HV**2**, and a low voltage source **32***b* and a low voltage drain **33***b* are formed in the low voltage device area LV**2**. Next, as shown in FIG. **2**P, for example, using a silicidation metal process step, plural silicide metal layers **34** are simultaneously formed corresponding to the upper surfaces of the high voltage gate **30***a*, a high voltage source **32***a*, a high voltage drain **33***a*, a low voltage gate **30***b*, a low voltage source **32***b*, a low voltage drain **33***b*, the first electrode plate **26***b*, and the second electrode plate **30***c*. Subsequently, as shown in FIG. **2**Q, an inter-layer dielectric (ILD) layer **35** is formed on the substrate **21**, completely covering the high voltage gate **30***a*, the low voltage gate **30***b*, and the second electrode plate **30***c*. As shown in FIG. **2**Q, plural electrical contact plugs **36** are then formed (for example, penetrating) in the ILD layer **35** to electrically connect the multiple silicide metal layers **34**.

[0033] In summary, the present invention utilizes high voltage polysilicon hard masks and the first electrode plate to achieve two-stage inclined high voltage bottom thermal oxide regions and high voltage CVD oxide regions without additional photomasks, simultaneously forms a PIP capacitor, and avoids shallow trench isolation (STI) notching, while reducing thermal processes to prevent exceeding the thermal budget during integration.

[0034] The present invention has been described in considerable detail with reference to certain preferred embodiments thereof. It should be understood that the description is for illustrative purpose, not for limiting the broadest scope of the present invention. Those skilled in this art can readily conceive variations and modifications within the spirit of the present invention. The various embodiments described above are not limited to being used alone; two embodiments may be used in combination, or a part of one embodiment may be used in another embodiment. For example, other process steps or structures, such as a deep well region, may be added. For another example, the lithography process step is not limited to the mask technology but it can also include electron beam lithography, immersion lithography, etc. Therefore, in the same spirit of the present invention, those skilled in the art can think of various equivalent variations and various combinations, and there are many combinations thereof, and the description will not be repeated here. The scope of the present invention should include what are defined in the claims and the equivalents.

What is claimed is:

1. A manufacturing method of an integrated structure of a semiconductor integrated structure having high and low voltage devices and a capacitor, comprising:

forming a bottom thermal oxide layer on a substrate, wherein the bottom thermal oxide layer completely covers a high voltage device area, a low voltage device area, and a capacitor area of the substrate;

forming a high voltage well in the substrate of the high voltage device area;

forming a chemical vapor deposition (CVD) oxide layer that completely covers the bottom thermal oxide layer;

forming a polysilicon hard mask layer that completely covers the CVD oxide layer;

etching the polysilicon hard mask layer to simultaneously form a high voltage polysilicon hard mask in the high voltage device area and a first electrode plate in the capacitor area;

etching the CVD oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form a high voltage CVD oxide region in the high voltage device area and a capacitor CVD oxide region in the capacitor area;

forming a low voltage well in the substrate of the low voltage device area;

etching the bottom thermal oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form a high voltage bottom thermal oxide region in the high voltage device area and a bottom thermal oxide region in the capacitor area;

forming a high voltage gate oxide layer over the high voltage device area on the substrate;

forming a low voltage gate oxide layer over the low voltage device area on the substrate;

when forming the high voltage gate oxide layer or the low voltage gate oxide layer, simultaneously forming a capacitor dielectric layer, connected and fully covering the first electrode plate;

forming a gate polysilicon layer, connected and fully covering the high voltage gate oxide layer, the low voltage gate oxide layer, and the capacitor dielectric layer; and

etching the gate polysilicon layer to simultaneously form a high voltage gate in the high voltage device area, a low voltage gate in the low voltage device area, and a second electrode plate in the capacitor area.

2. The manufacturing method of claim **1**, wherein the step of forming the low voltage well in the substrate of the low voltage device area includes: using an ion implantation process step, with the bottom thermal oxide layer serving as a sacrificial layer, and accelerating ions to penetrate the sacrificial layer to implant into the low voltage device area to form the low voltage well.

3. The manufacturing method of claim **1**, wherein the step of etching the CVD oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form the high voltage CVD oxide region in the high voltage device area and the capacitor CVD oxide region in the capacitor area includes: using a wet etching process step to etch the CVD oxide layer.

4. The manufacturing method of claim **1**, wherein the step of etching the bottom thermal oxide layer using the high voltage polysilicon hard mask and the first electrode plate as etching barriers to simultaneously form the high voltage bottom thermal oxide region in the high voltage device area and the bottom thermal oxide region in the capacitor area includes: using a wet etching process step to etch the bottom thermal oxide layer, such that the high voltage side wall of the high voltage bottom thermal oxide region and the capacitor side wall of the bottom thermal oxide region each have an inclined angle with respect to the upper surface of the substrate.

5. The manufacturing method of claim **1**, further comprising: after forming the polysilicon hard mask layer, using an ion implantation process step to accelerate N-type or P-type ions to implant into the polysilicon hard mask layer.

6. The manufacturing method of claim **1**, further comprising: after forming the gate polysilicon layer, using an ion implantation process step to accelerate N-type or P-type ions to implant into the gate polysilicon layer.

7. The manufacturing method of claim **1**, wherein the high voltage gate is in direct contact with the high voltage polysilicon hard mask.

8. The manufacturing method of claim **1**, further comprising: after forming the high voltage gate, the low voltage gate, and the second electrode plate, forming two high voltage spacers corresponding to and connected on either side of the high voltage gate, and two low voltage spacers corresponding to and connected on either side of the low voltage gate.

9. The manufacturing method of claim **8**, further comprising: after forming the high voltage gate, the low voltage gate, and the second electrode plate, forming two capacitor spacers corresponding to and connected on either side of the second electrode plate.

10. The manufacturing method of claim **1**, further comprising: simultaneously forming a high voltage source and a high voltage drain in the high voltage device area, and a low voltage source and a low voltage drain in the low voltage device area.

11. The manufacturing method of claim **10**, further comprising: using a silicidation metal process step to simultaneously form a plurality of silicide metal layers corresponding to the upper surfaces of the high voltage gate, the high voltage source, the high voltage drain, the low voltage gate, the low voltage source, the low voltage drain, the first electrode plate, and the second electrode plate.

12. The manufacturing method of claim **1**, further comprising: forming an inter-layer dielectric (ILD) layer on the substrate, completely covering the high voltage gate, the low voltage gate, and the second electrode plate.

13. The manufacturing method of claim **12**, further comprising: forming a plurality of electrical contact plugs in the ILD layer, electrically connecting the multiple silicide metal layers.

14. The manufacturing method of claim **1**, further comprising: forming a high voltage body region in the high voltage well of the high voltage device area, wherein the high voltage source is located in the high voltage body region.

* * * * *